B020313(020)

B.Tech. (Third Semester) Examination Nov.-Dec. 2022

(Civil Engineering Branch)

INTRODUCTION to SOLID MECHANICS

Time Allowed : Three hours

Maximum Marks : 100

Note : Attempt all questions. Part (a) of each question is compulsory and carries 4 marks. Attempt any two parts from (b), (c), (d) which carry 8 marks each.

Unit-I

1. (a) Explain stress-strain graph of mild steel. 4

(b) For the steel bar shown in figure, determine the longitudinal force P and stress p at all cross-sections.

Determine also the vertical displacement Δ of all cross-sections of the bar. Represent the result graphically by plotting P, p and Δ diagrams. Take $E = 2 \times 10^5$ N/mm^2. Also locate the section of zero displacement. 8

(c) The aluminium and steel pipes shown in figure are fastened to rigid supports at one of their ends and to a rigid plate C at the other ends. Derive expressions for axial stresses in the two pipes. Hence find the numerical values if $P = 30$ kN, $A_a = 4000$ mm^2, $As = 400$ mm^2, $Ea = 0{\cdot}7 \times 10^5$ N/mm^2 and $Es = 2 \times 10^5$ N/mm^2. 8

(d) Derive the relation between shear modulus, modulus of elasticity and Young's modulus. 8

Unit-II

2. (a) Define principal stresses, principal plane and Mohr circle. 4

(b) Derive the equation of stresses due to pure shearing. 8

(c) A short metallic column of 500 mm^2 cross-sectional area an axial compressive load of 100 kN. For a plane inclined at 60° with the direction of load, calculate : 8

(i) Normal stress,

(ii) Tangential stress,

(iii) Resultant stress,

(iv) Maximum shear stress and

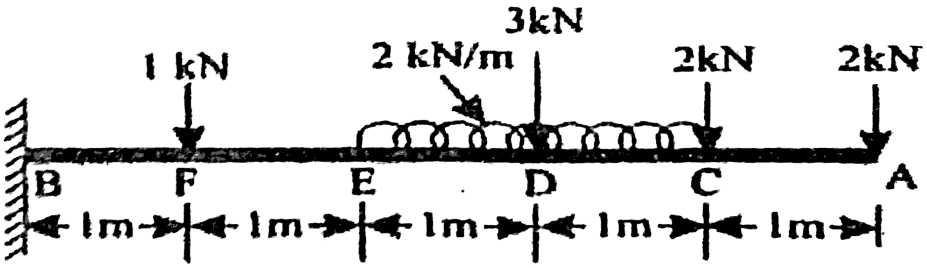
(v) Obliquity of the resultant stress.

(d) A point is subjected to perpendicular stresses of 50 MN/m^2 and 30 MN/m^2 both tensile. Calculate the normal, tangential stresses and resultant stress and its obliquity on a plane making an angle of 30° with the axis of second stress. 8

Unit-III

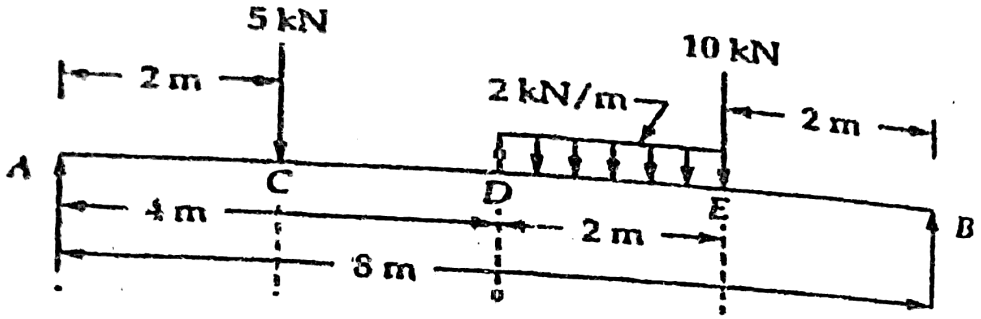
3. (a) What do you mean by pure bending? Explain the use of shear force and bending moment diagram. 4

(b) Draw the shear force and bending moment diagram for given cantilever beam. 8



(c) Derive the relation between load, shear force and bending moment. 8

(d) Draw the shear force and bending moment diagram for given beam. 8



Unit-III

4. (a) Write the assumptions and limitations of Euler's formula. 4

(b) Derive the expression for Euler's load for a column with one end fixed and other end free. 8

(c) Compare the crippling loads given by Euler's and Rankine's formulae, for a tubular steel strut 2·5 m long, having outer and inner diameter as 40 mm and 30 mm respectively, loaded through pin joints at the ends. Take yield stress as 320 N/mm^2. The Rankin's constant = 1/7500 and $E = 2 \times 10^5$ N/mm^2, for

what length of the strut of this cross-section does the Euler's formula cease to apply? 8

(d) A hollow cast iron column with fixed ends supports an axial of 800 kN. If the column is 3 m long and has an external diameter of 200 mm, find the thickness of metal required. Use Rankine's formula, taking a constant of 1/6400 and assume a working stress of 90 N/mm^2. 8

Unit-V

5. (a) Determine the power transmitted by a solid shaft of 5 cm diameter while rotating at 120 rpm. Maximum permissible shear stress is 8 kN/cm^2. 4

(b) A solid of 10 cm diameter transmits 74 kW at 150 rev/min. Calculate (i) The torque on the shaft, (ii) the maximum shear stress developed, (c) the angle of twist in a length of 1·50 m, and (iv) the shear stress at a radius of 3 cm. Take $C = 8$ MN/cm^2. 8

(c) A hollow steel shaft with a length of 1.8 m is loaded by torque of 6 kNm. Determine the external and internal diameter of the shaft, if the angle of twist is not to exceed 2°, and the shear stress is not to

exceed 7 kN/cm^2. The modulus of rigidity of steel is 8 MN/cm^2. 8

(d) A closed-coiled helical spring of 10 cm mean diameter is made of a 1 cm diameter rod and has 20 turns. The spring carries an axial load of 200 N. Determine the shearing stress and the deflection developed in the spring. Also, find the stiffness of the spring and frequency of its vibrations. Take $C = 84\ GN/m^2$. 8